Lockout/Tagout (LOTO) Procedure

Purpose and Scope

This procedure establishes the minimum requirements for isolating hazardous energy before any servicing, maintenance, cleaning, or unjamming activity on Riverbend Packaging equipment. It applies to the filling and capping line, all conveyors, the RP-200 centrifugal pump, the CX-4 capper, and the hydraulic press. It applies to every employee and contractor who services equipment where unexpected startup or release of stored energy could cause injury.

Hazardous energy includes electrical, hydraulic, pneumatic, mechanical (spring or gravity), thermal, and chemical sources. Normal production operations using the machine's own guarded controls are not covered, but any task that requires removing a guard, bypassing an interlock, or placing any part of the body in a danger zone is covered.

WARNING: Pressing the stop button or opening a door interlock is NOT lockout. Only a personally applied lock and tag on an energy-isolating device, followed by verification, protects you.

Roles and Responsibilities

Authorized Employees

Authorized employees are trained to apply locks and perform servicing. Each authorized employee owns one red lock with a unique key and a personal tag showing name, date, and department. Locks are never shared and keys never left in the lock.

Affected Employees

Affected employees operate or work near locked-out equipment. They must never attempt to restart, remove a lock, or operate a tagged control.

EHS Lead

The EHS Lead audits the program annually, maintains the energy-source inventory, and authorizes lock removal when the owner is unavailable.

Lockout Sequence

Follow every step in order. Do not skip steps for short tasks.

1. Notify all affected employees that servicing requires a shutdown and identify the machine and energy types.
2. Shut the machine down using the normal stop procedure.
3. Locate and operate each energy-isolating device (main disconnect, hydraulic supply valve, air supply valve, product valve).
4. Apply your personal lock and tag to each isolating device. Use a hasp when more than one person works on the machine.
5. Relieve or restrain stored energy: bleed air lines to zero, lower or block raised loads, release hydraulic accumulators, and allow hot components to cool.

6 Verify isolation. Attempt to start the machine from the local control station and confirm nothing moves, then return controls to OFF. Use a meter to confirm zero voltage on electrical work.

7 Begin work only after verification passes.

Energy Source Inventory

Equipment	Energy Types	Isolation Point	Location
Filler/Capper line	Electrical, pneumatic	Main disconnect MD-1, air valve AV-1	Line 1 panel
Conveyors C1-C4	Electrical, gravity (inclines)	Disconnects MD-2 through MD-5	Conveyor MCC
RP-200 pump	Electrical, product pressure	Disconnect MD-6, product valves PV-1/PV-2	Pump room
CX-4 capper	Electrical, pneumatic, spring (clutch)	MD-7, AV-2	Capper base
Hydraulic press	Electrical, hydraulic, gravity (ram)	MD-8, HV-1, ram block	Press bay

Restoring Equipment to Service

1 Remove tools and replace all guards and interlocks.

2 Confirm all personnel are clear of the danger zone.

3 Check that controls are in the OFF or neutral position.

4 Remove each lock and tag. Only the person who applied a lock may remove it.

5 Notify affected employees that the machine is being re-energized.

6 Re-energize and run a function check before returning to production.

Group Lockout

When several people service one machine, the lead authorized employee places a lock on the isolating device and a lockbox. Every worker then places a personal lock on the lockbox, which holds the keys to the primary locks. Each person removes their own lock when their work is finished. The machine cannot be energized until the last lock is removed.

Shift Change and Lock Removal

If work continues across shifts, the outgoing authorized employee hands over in person, and the incoming employee applies a lock before the outgoing lock is removed. If an employee leaves the plant with a lock still applied, the EHS Lead may authorize removal only after attempting to contact the employee, verifying they are not on site, and inspecting the machine. The employee must be informed before they resume work.

WARNING: Never cut a lock to restart equipment without EHS Lead authorization and a documented search for the lock owner.

Training and Audit

Authorized employees complete initial training and retraining after any procedure change or observed violation. Periodic inspections are performed annually by the EHS Lead and documented with the machine, observed employee, and findings. Records are retained for three years.

Common Mistakes

- Locking only the control panel rather than the main disconnect.
- Forgetting stored energy such as a raised press ram or pressurized air line.
- Skipping the verification attempt because "the light is off".
- Leaving a tag on without a lock where a lock could have been applied.
- Using a lock that is not personally assigned.